

TECHNICAL DOCUMENTATION

Instructor Performance and Course Quality Evaluation

A data-driven framework for measuring teaching effectiveness, course consistency, and instructor influence across the EduPro learning platform.

PROJECT PURPOSE	PRIMARY USERS
Replace fragmented, subjective evaluation with repeatable instructor and course quality measurement.	EduPro leadership, academic operations, course owners, and public-sector education stakeholders.
ANALYTICAL FOCUS	DELIVERABLES
Instructor ratings, experience, expertise, course ratings, and enrollment activity.	EDA research paper, Streamlit dashboard specification, and executive stakeholder summary.

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1. Executive summary

Decision to support. EduPro needs a common evidence base for recognizing excellent teaching, detecting inconsistent course quality, and targeting instructor development where it will most improve learner outcomes.

The framework integrates teacher, course, and transaction data to measure whether instructional expertise and experience are associated with stronger learner-facing course ratings and enrollment demand. It is designed to support fair comparison while keeping the limitations of observational data visible.

The resulting dashboard should be used for continuous improvement and quality assurance, not as the sole input to employment, promotion, or disciplinary decisions. Ratings must be interpreted alongside course volume, category mix, learner feedback, and sample size.

Identify excellence

Rank instructors using performance, consistency, and sufficient-volume rules.

Diagnose gaps

Surface low-rated or highly variable course experiences by category and level.

Target action

Prioritize coaching, peer review, and course redesign where impact is highest.

2. Problem statement and scope

Online-learning quality depends heavily on instructor effectiveness and course design. Without a structured framework, EduPro cannot reliably answer which instructors consistently deliver high-quality experiences, whether experience is linked to ratings, which categories depend most on teaching quality, or how performance is distributed across the platform.

In scope: instructor profiles, course quality, rating relationships, expertise comparisons, and transaction-based enrollment proxies.

Out of scope: causal claims about instructor impact, learner-level segmentation, compensation decisions, and automated high-stakes personnel action.

3. Data model and integration

Use three source sheets. Keep source-level identifiers intact and create a documented analytical dataset at the transaction grain: one row per enrollment/transaction linked to a course and instructor.

Source	Key fields	Analytical role
Teachers	TeacherID, TeacherName, Age, Gender, Expertise, YearsOfExperience, TeacherRating	Instructor attributes and baseline teaching-quality score
Courses	CourseID, CourseName, CourseCategory, CourseLevel, CourseRating	Course design and learner-facing quality outcome
Transactions	TransactionID, CourseID, TeacherID	Enrollment volume and teacher-course mapping validation

Integration controls. Test null IDs, duplicate primary keys, orphaned transactions, teacher-course conflicts, and courses mapped to multiple teachers. Report the retained-row percentage and each exclusion reason before drawing

conclusions.

4. Analytical methodology

1. Prepare

Standardize IDs, rating scales, category labels, and experience values. Create missing-data flags rather than silently dropping records.

2. Profile

Describe instructor age, gender, expertise, experience, teacher ratings, course ratings, and enrollment distribution.

3. Relate

Calculate Pearson and Spearman correlations for experience versus ratings; inspect scatter plots and non-linear trend lines.

4. Compare

Group results by expertise, course category, course level, and instructor rating tier. Include sample counts and confidence context.

5. Act

Translate findings into quality interventions, monitoring thresholds, and a periodic review cadence.

5. KPIs and calculation rules

KPI	Definition	Interpretation
Average Teacher Rating	Mean TeacherRating; report median and n alongside it.	Platform teaching-quality benchmark.
Average Course Rating	Mean CourseRating, optionally transaction-weighted.	Content and delivery effectiveness.
Rating Consistency Index	1 - normalized within-instructor course-rating standard deviation. Show only where an instructor has at least 3 rated courses.	Reliability of experience across courses.
Experience Impact Score	Standardized regression coefficient or correlation of YearsOfExperience with rating, with p-value and n.	Directional association, not causation.
Enrollment Influence Ratio	Average transaction count for high-rated instructor tier divided by average count for all other tiers.	Association between instructor rating and demand.

Suggested rating tiers: Low < 3.5, Mid 3.5-4.2, High > 4.2 on a five-point scale. Confirm these thresholds against EduPro's actual rating policy and distribution before publication.

6. Streamlit dashboard specification

The dashboard should default to an all-platform view and allow users to narrow the population without changing the underlying definitions.

Module	Primary visual	Decision supported
Instructor leaderboard	Sortable table: teacher rating, average course rating, consistency, courses, enrollments	Who is consistently strong, and where is review needed?
Experience vs. rating	Scatter plot with trend line and correlation annotation	Does experience relate to instructional performance?
Course quality matrix	Heatmap: course category x level, colored by average course rating	Which offerings show systematic quality variation?
Expertise comparison	Ranked bars / box plots by expertise area	Which domains are strongest and most variable?
Enrollment influence	Tier comparison bars with counts and volume context	Are highly rated instructors associated with demand?

Controls: expertise multiselect, course category selector, course level selector, teacher-rating range slider, and minimum enrollment/sample-size filter. Every chart should display the active population count and expose the underlying rows for export.

7. Insight framework and interpretation

The final EDA should answer the following questions using both summary statistics and visual evidence:

- How concentrated or dispersed are instructor ratings, and are there meaningful outliers?
- Does experience show a linear, threshold, diminishing-return, or no observable relationship with teacher and course ratings?
- Do high-rated instructors also have higher-rated courses after category and level are considered?
- Which expertise areas have both strong average quality and low variability?
- Are enrollment differences robust once course availability and catalog mix are considered?

Interpretation guardrails. Small samples can produce volatile ratings. Category, course level, price, recency, promotion, and learner mix may confound observed relationships. Always show n, avoid ranking instructors with insufficient volume, and label correlation as association.

8. Recommendations and governance

Priori ty	Recommended action	Evidence trigger
1	Establish a quarterly quality review using the KPI definitions in this document.	Dashboard flags, trend shifts, or category-level declines.
2	Recognize sustained excellence through peer mentoring and reusable teaching practices.	High rating plus consistency and minimum course/enrollment volume.
3	Offer developmental support before punitive action.	Low course ratings, high variability, or a persistent gap versus comparable category peers.
4	Commission course-design reviews in weak category-level cells.	Low category x level averages with adequate sample size.
5	Review fairness and privacy controls before broader rollout.	Any use beyond improvement, reporting, and curriculum planning.

9. Delivery checklist

- Research paper: data quality log, EDA visuals, tested relationships, insights, limitations, and prioritized recommendations.
- Streamlit dashboard: data refresh method, filters, KPI definitions, downloadable tables, and clear empty-state handling.
- Executive summary: concise findings, recommended actions, risks, owners, and quarterly measures of progress.
- Validation: reconcile row counts, test filter behavior, verify metric calculations, and complete stakeholder review before publication.

This framework shifts evaluation from isolated opinions to transparent, repeatable evidence - enabling EduPro to improve teaching quality while using instructor data responsibly.